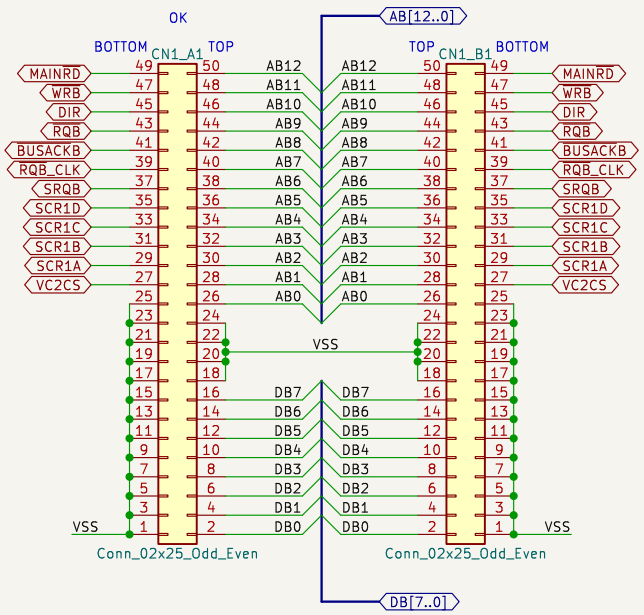
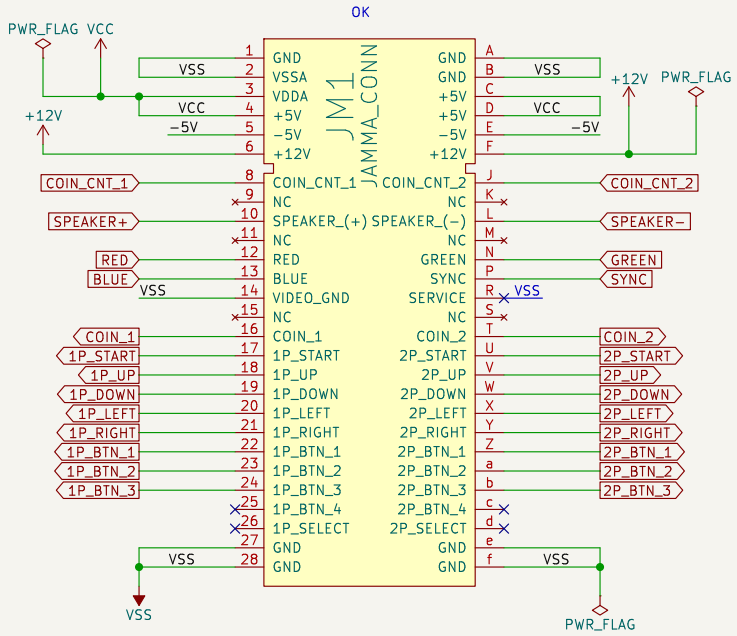
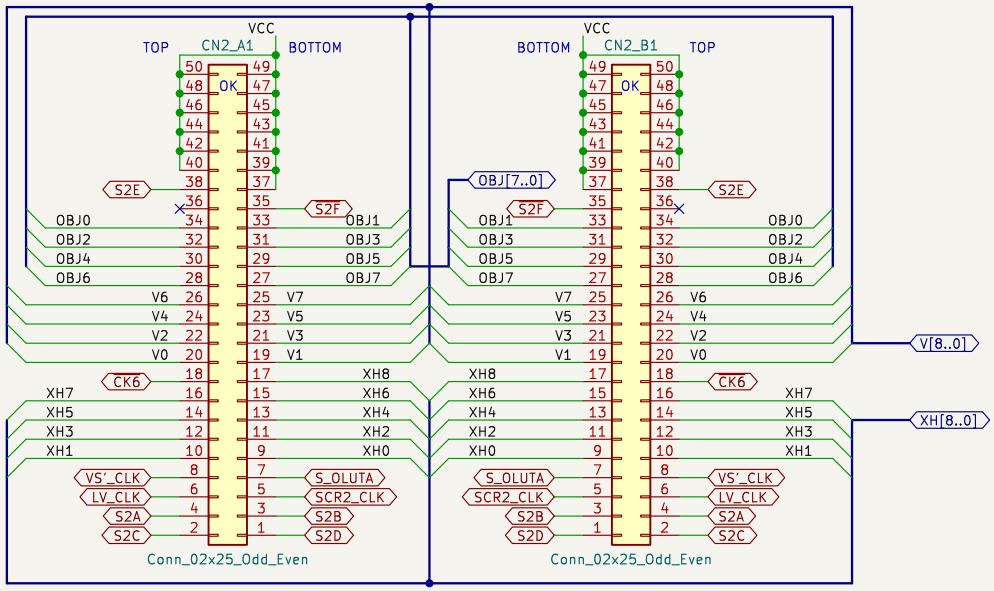






Observation
The pcb that has the jamma connector is the upper level and the location of each device is labeled with number, letter and number1. On the lower level pcb, the devices are labeled with letter and number.

video
File: video.kicad_sch

sound
File: sound.kicad_sch

main
File: main.kicad_sch

interruption
File: interruption.kicad_sch

video_timings
File: video_timings.kicad_sch

v_counter
File: v_counter.kicad_sch

h_counter
File: h_counter.kicad_sch

last_u
File: last_u.kicad_sch

capacitors
File: capacitors.kicad_sch

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JOTEGO
Sheet: /
File: exed.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 1/28

v_scroll

File: v_scroll.kicad_sch

char

File: char.kicad_sch

h_scroll

File: h_scroll1.kicad_sch

object_table

File: object_table.kicad_sch

scroll1

File: scroll1.kicad_sch

obj_ram

File: obj_ram.kicad_sch

scroll2

File: scroll2.kicad_sch

obj_lut

File: obj_lut.kicad_sch

colmix

File: colmix.kicad_sch

obj_counters

File: obj_counters.kicad_sch

logics_gates

File: logics_gates.kicad_sch

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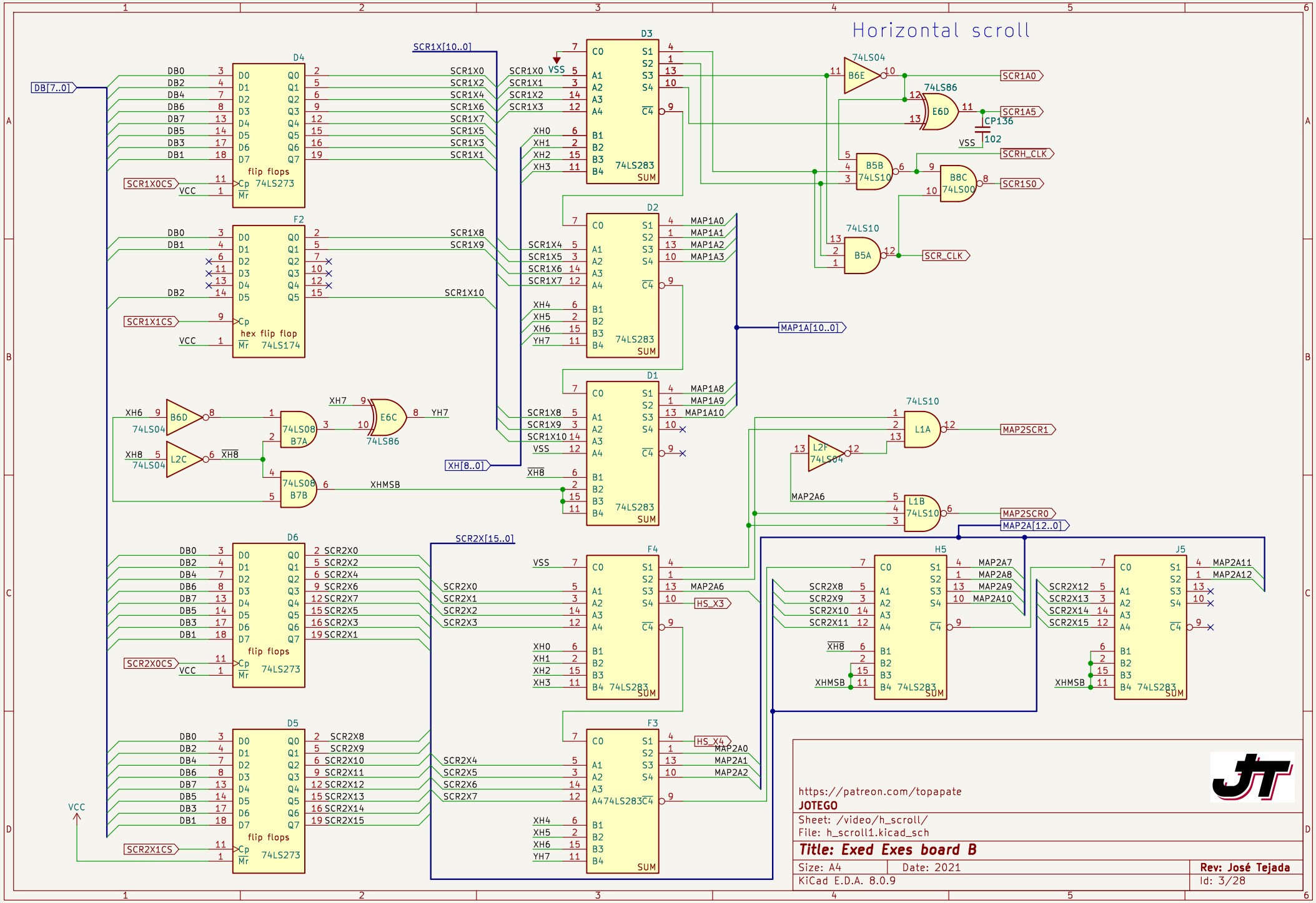
JOTEGO

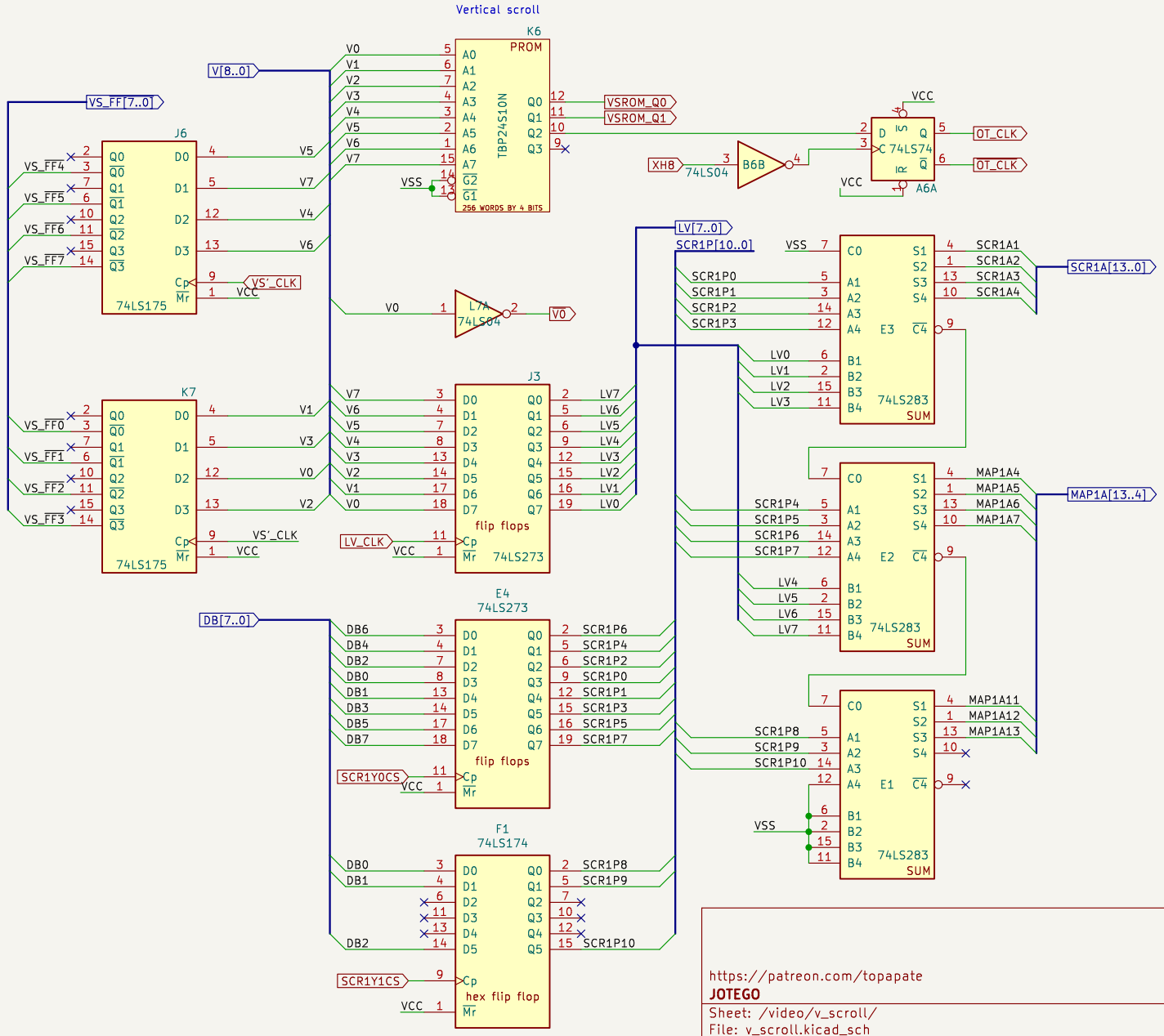
Sheet: /video/

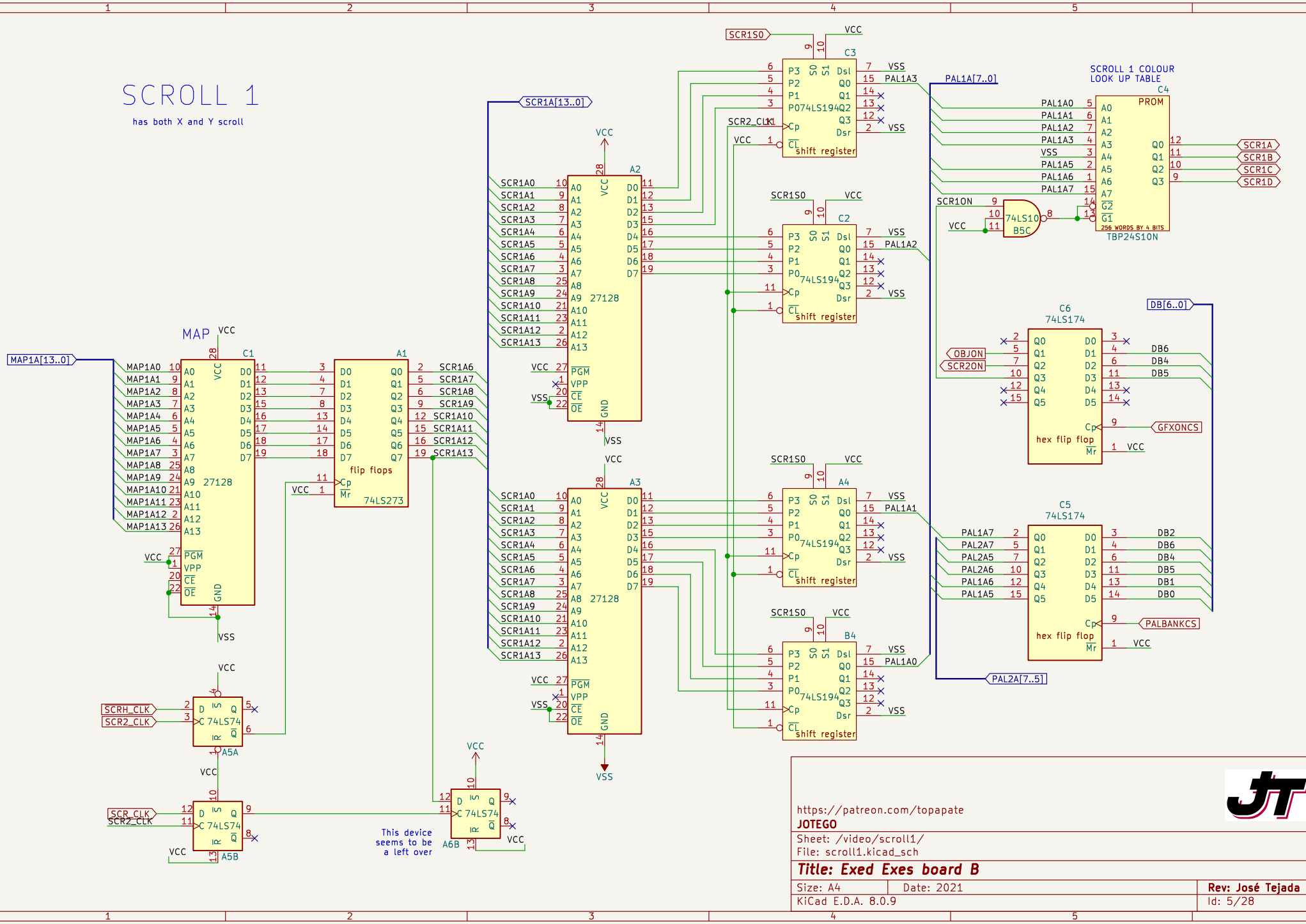
File: video.kicad_sch

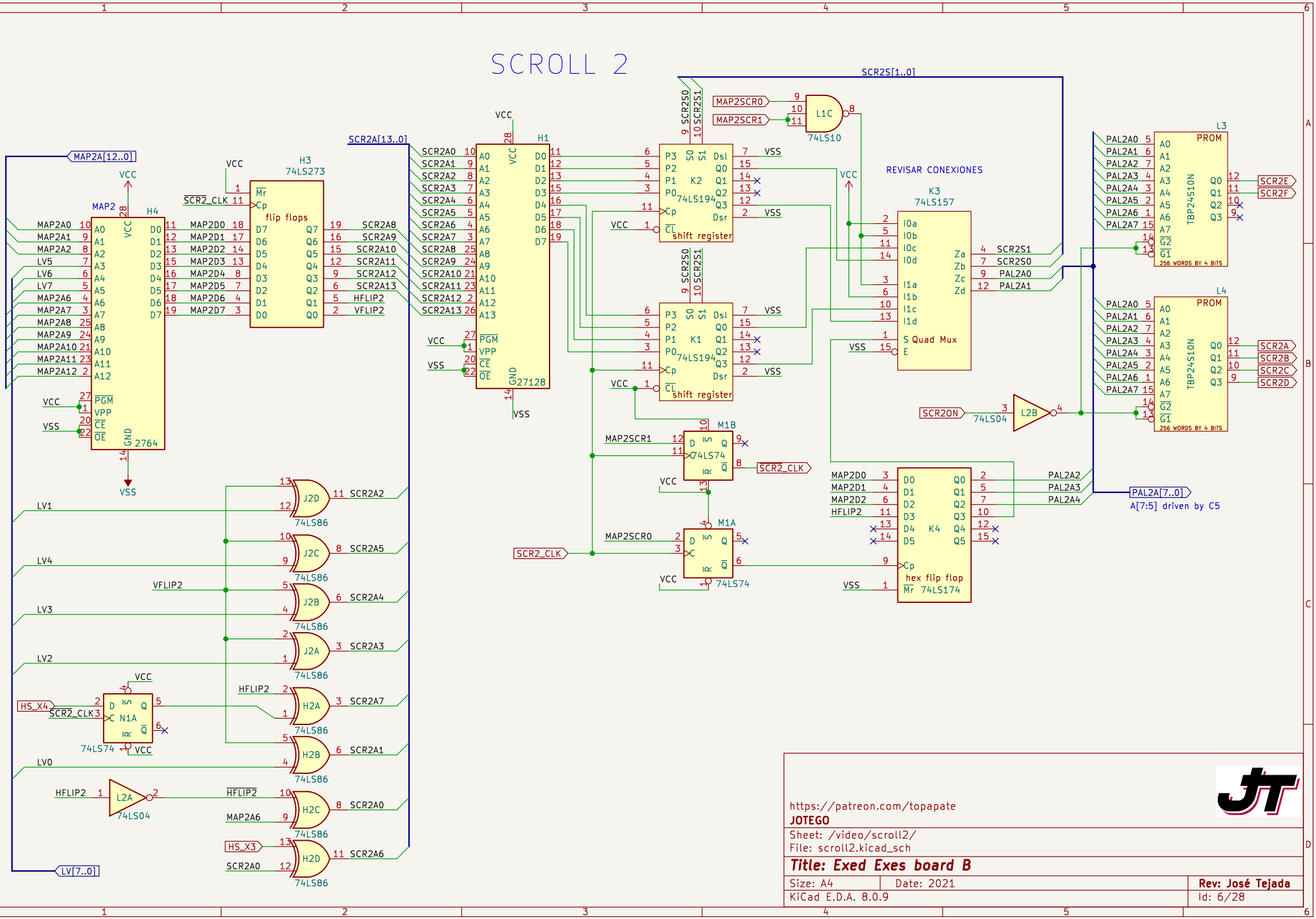
Title: Exed Exes board B

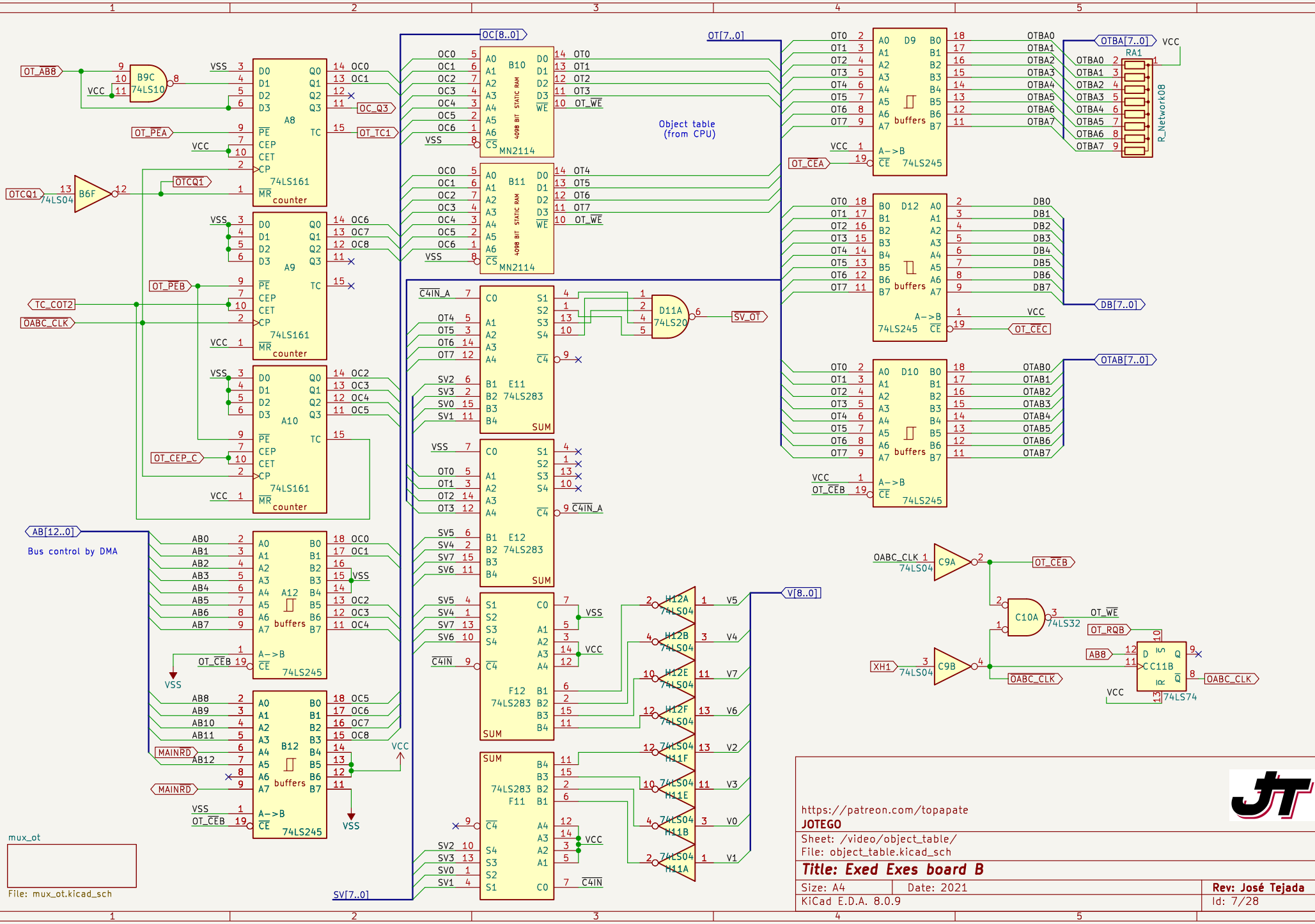
Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 2/28

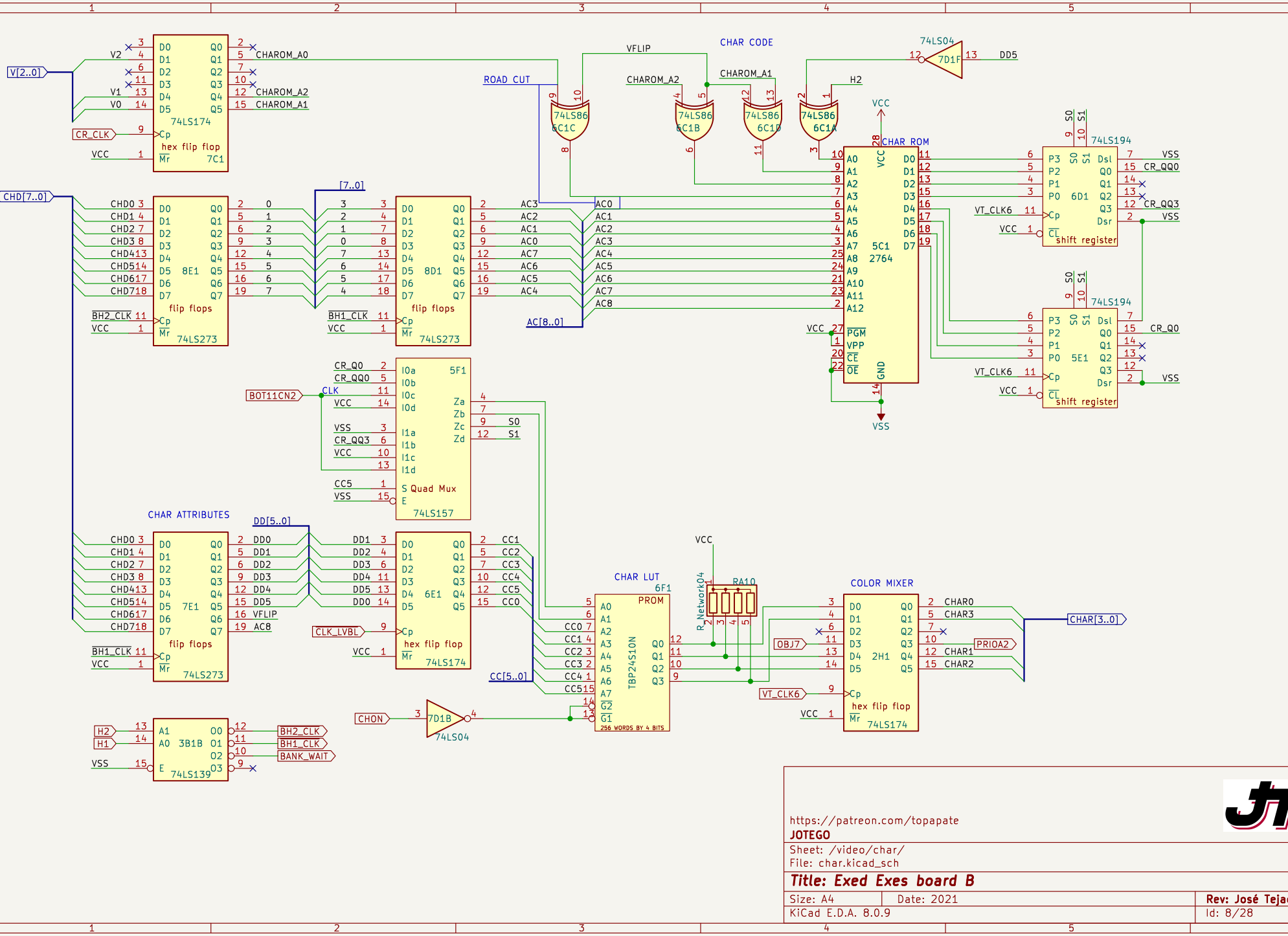

Horizontal scroll
https://patreon.com/topapate
JOTEGO
Sheet: /video/h_scroll/
File: h_scroll1.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
Rev: José Tejada
KiCad E.D.A. 8.0.9
Id: 3/28


Vertical scroll
K6
PROM
TBP24S10N
256 WORDS BY 4 BITS
V[8..0]
VS_FF[7..0]
VSROM_Q0
VSROM_Q1
J6
74LS175
VS_FF4
VS_FF5
VS_FF6
VS_FF7
VS'_CLK
VCC
VSS
XH8
74LS04
B6B
A6A
74LS74
OT_CLK
K7
VS_FF0
VS_FF1
VS_FF2
VS_FF3
J3
flip flops
74LS273
LV_CLK
LV[7..0]
SCR1P[10..0]
74LS283
SUM
SCR1A[13..0]
MAP1A[13..4]
E4
DB[7..0]
SCR1Y0CS
F1
74LS174
SCR1Y1CS
hex flip flop
https://patreon.com/topapate
JOTEGO
Sheet: /video/v_scroll/
File: v_scroll.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
KiCad E.D.A. 8.0.9
Rev: José Tejada
Id: 4/28


SCROLL 1
has both X and Y scroll
MAP
MAP1A[13..0]
C1
27128
A1
74LS273
flip flops
SCR1A[13..0]
A2
A3
C3
C2
A4
B4
74LS194
shift register
SCR1S0
SCR2_CLK
SCR1A
SCR1B
SCR1C
SCR1D
SCROLL 1 COLOUR
LOOK UP TABLE
C4
PROM
TBP24S10N
256 WORDS BY 4 BITS
PAL1A[7..0]
SCR1ON
74LS10
B5C
C6
74LS174
hex flip flop
OBJON
SCR2ON
DB[6..0]
GFXONCS
C5
PALBANKCS
PAL2A[7..5]
SCRH_CLK
SCR2_CLK
74LS74
A5A
SCR_CLK
SCR2_CLK
A5B
A6B
This device
seems to be
a left over
https://patreon.com/topapate
JOTEGO
Sheet: /video/scroll1/
File: scroll1.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
KiCad E.D.A. 8.0.9
Rev: José Tejada
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SCROLL 2
SCR2S[1..0]
MAP2A[12..0]
SCR2A[13..0]
REVISAR CONEXIONES
MAP2
H4
2764
H3
74LS273
flip flops
H1
27128
K3
74LS157
S Quad Mux
L3
L4
PROM
TBP24S10N
256 WORDS BY 4 BITS
K4
hex flip flop
74LS174
74LS194
shift register
74LS74
74LS86
74LS04
74LS10
M1A
M1B
L1C
L2A
L2B
J2A
J2B
J2C
J2D
H2A
H2B
H2C
H2D
SCR2_CLK
HS_X4
HS_X3
HFLIP2
VFLIP2
LV0
LV1
LV2
LV3
LV4
LV[7..0]
SCR2ON
PAL2A[7..0]
A[7:5] driven by C5
https://patreon.com/topapate
JOTEGO
Sheet: /video/scroll2/
File: scroll2.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
Rev: José Tejada
KiCad E.D.A. 8.0.9
Id: 6/28


OT_AB8
B9C
74LS10
VCC
VSS
A8
74LS161
counter
OT_PEA
OTCQ1
B6F
74LS04
A9
OT_PEB
TC_COT2
OABC_CLK
A10
OT_CEP_C
AB[12..0]
Bus control by DMA
A12
buffers
A->B
74LS245
OT_CEB
MAINRD
B12
mux_ot
File: mux_ot.kicad_sch
OC[8..0]
B10
B11
4096 BIT STATIC RAM
MN2114
OT[7..0]
Object table
(from CPU)
OT_WE
C4IN_A
D11A
74LS20
SV_OT
E11
74LS283
SUM
E12
F12
C4IN
SV[7..0]
V[8..0]
H12A
H12B
H12E
H12F
H11F
H11E
H11B
H11A
D9
D12
D10
OTBA[7..0]
OTAB[7..0]
DB[7..0]
RA1
R_Network08
OT_CEA
OT_CEC
C9A
C10A
74LS32
C9B
XH1
OT_RQB
C11B
74LS74
https://patreon.com/topapate
JOTEGO
Sheet: /video/object_table/
File: object_table.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
KiCad E.D.A. 8.0.9
Rev: José Tejada
Id: 7/28



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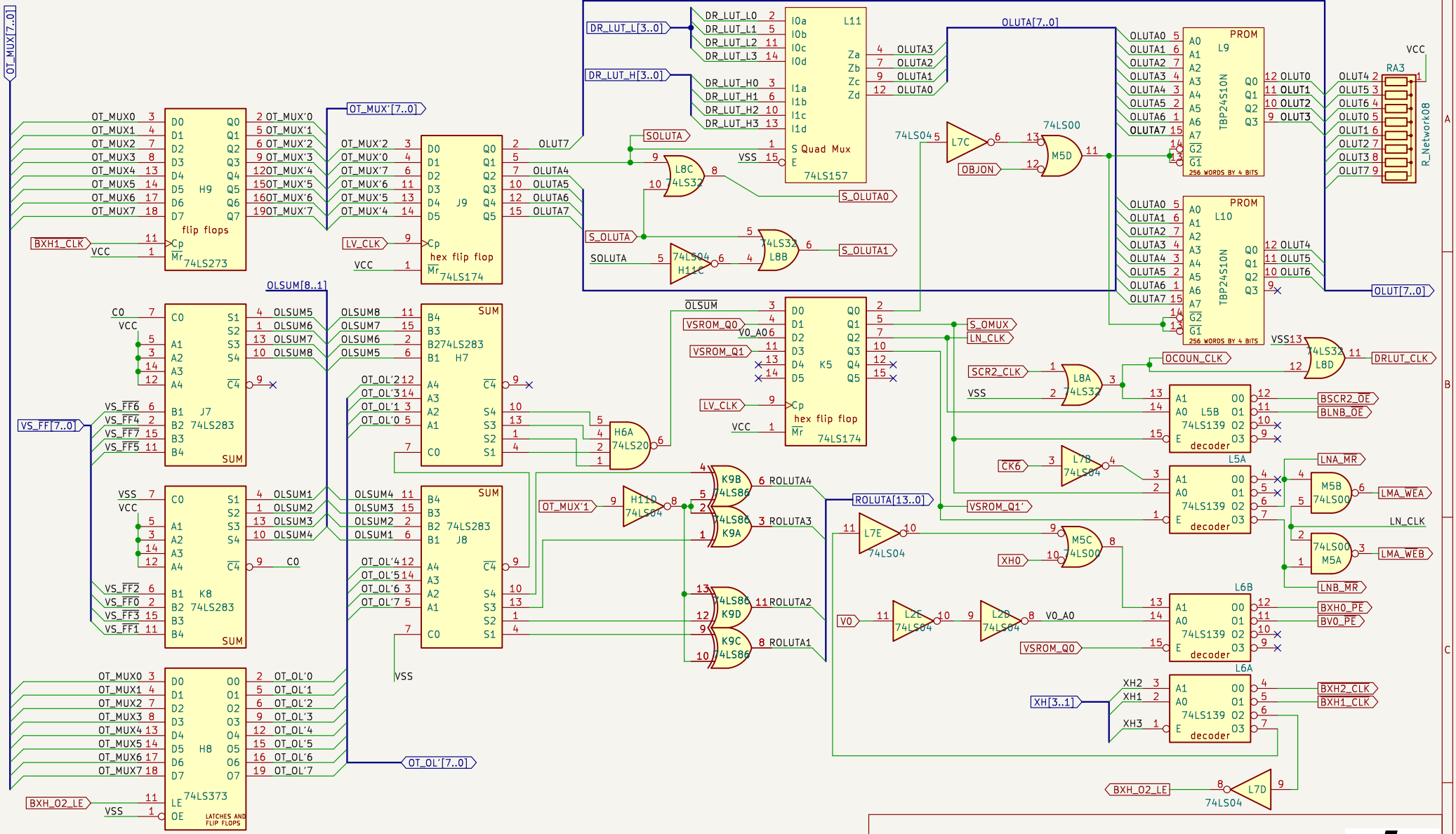
JOTEGO

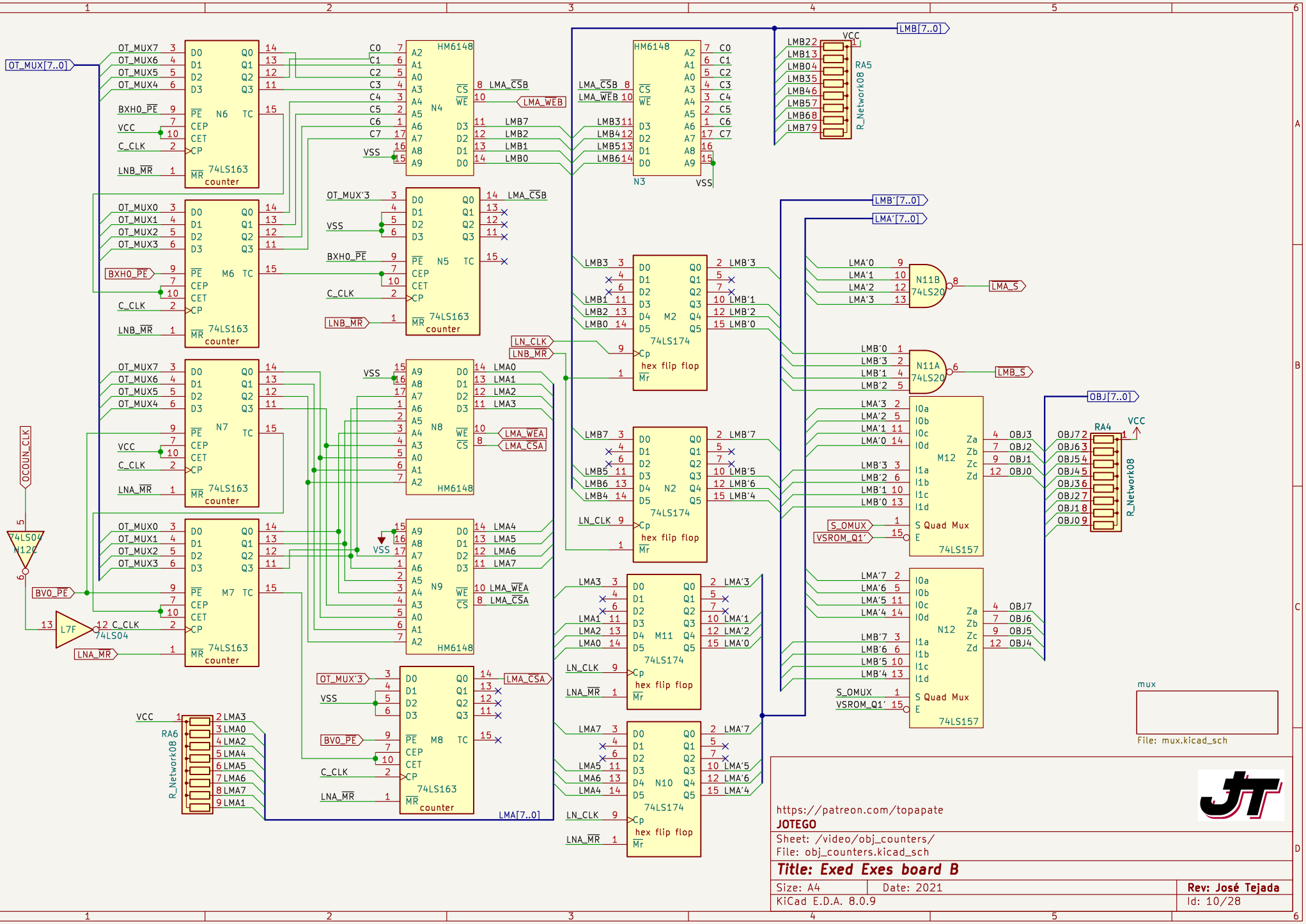
Sheet: /video/char/

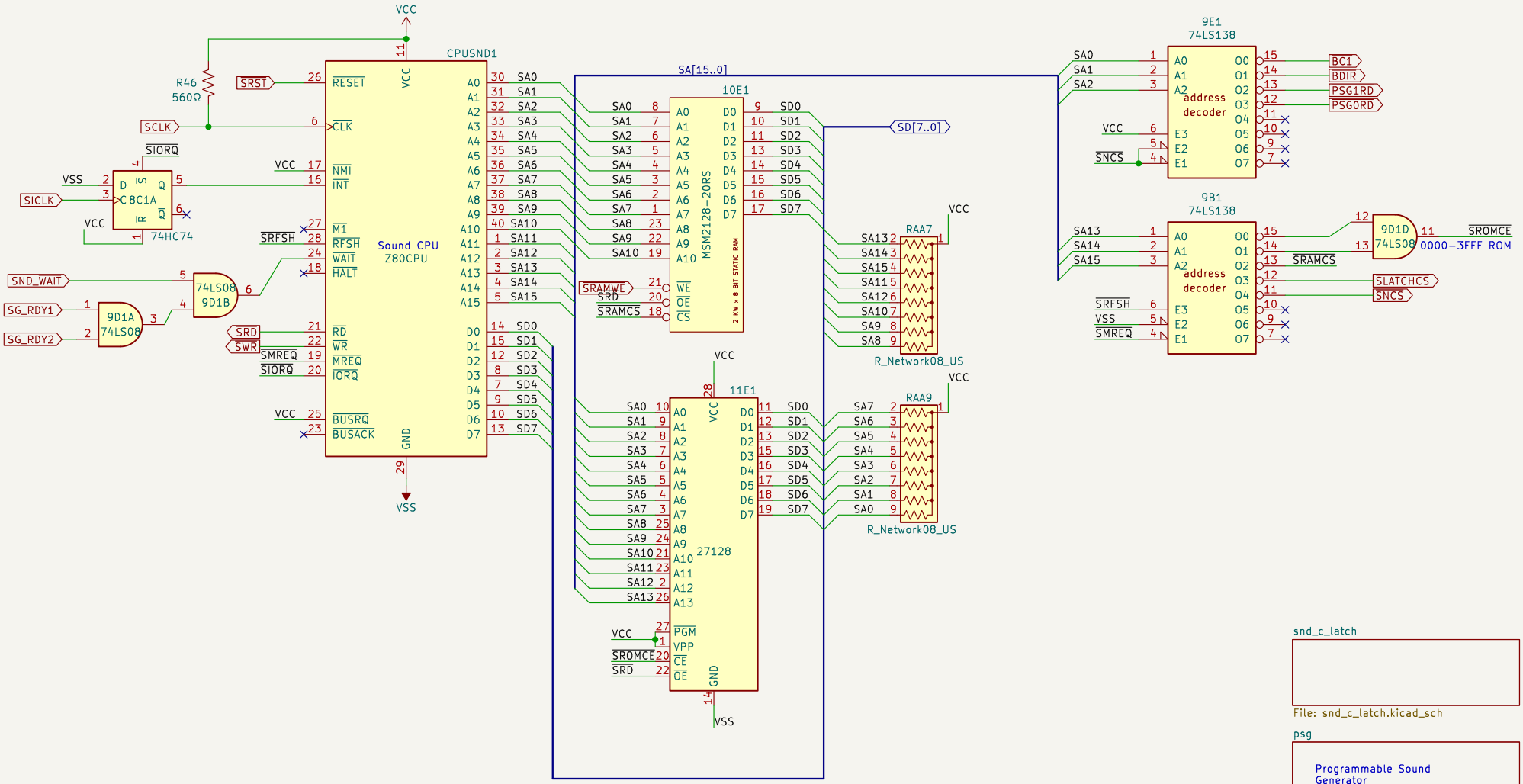
File: char.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 8/28


OT_MUX[7..0]
DR_LUT_L[3..0]
DR_LUT_H[3..0]
OLUTA[7..0]
OLUT[7..0]
OT_MUX'[7..0]
OLSUM[8..1]
VS_FF[7..0]
ROLUTA[13..0]
OT_OL'[7..0]
XH[3..1]
74LS157
74LS273
74LS174
74LS283
74LS373
74LS139
74LS32
74LS04
74LS00
74LS20
74LS86
TBP24S10N
R_Network08
flip flops
hex flip flop
decoder
LATCHES AND FLIP FLOPS
https://patreon.com/topapate
JOTEGO
Sheet: /video/obj_lut/
File: obj_lut.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
Rev: José Tejada
KiCad E.D.A. 8.0.9
Id: 9/28
rom_olut
File: rom_olut.kicad_sch


OT_MUX[7..0]
LMB[7..0]
LMB'[7..0]
LMA'[7..0]
OBJ[7..0]
LMA[7..0]
OCOUN_CLK
LN_CLK
LNB_MR
LNA_MR
LMA_CSB
LMA_WEB
LMA_WEA
LMA_CSA
LMA_S
LMB_S
S_OMUX
VSROM_Q1'
BXH0_PE
BV0_PE
C_CLK
VCC
VSS
N4
N5
N6
N7
N8
N9
N10
N11
N12
M2
M6
M7
M8
M11
M12
74LS163
74LS174
74LS157
74LS20
74LS04
HM6148
R_Network08
RA4
RA5
RA6
mux
File: mux.kicad_sch
https://patreon.com/topapate
JOTEGO
Sheet: /video/obj_counters/
File: obj_counters.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
Rev: José Tejada
KiCad E.D.A. 8.0.9
Id: 10/28



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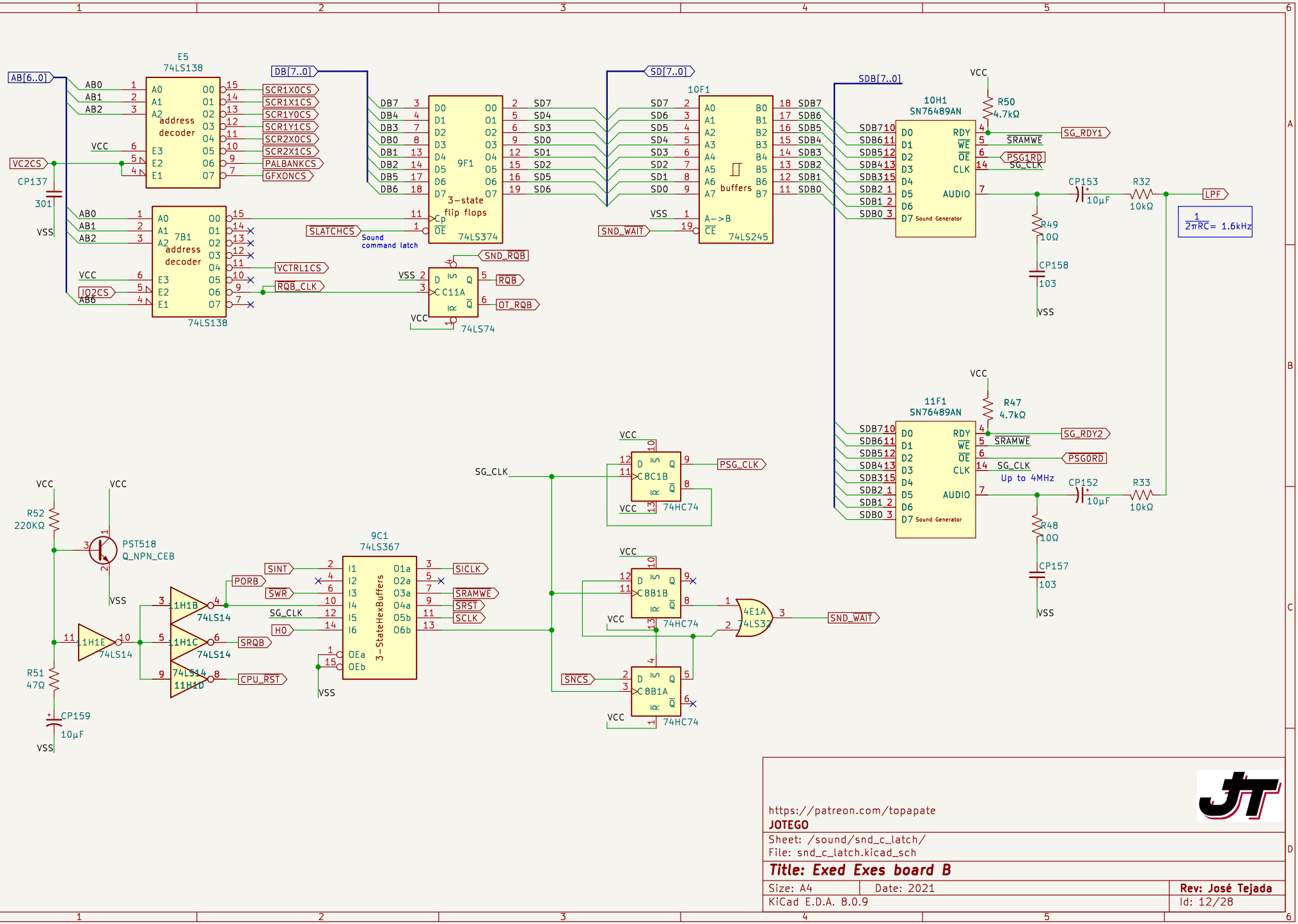
JOTEGO

Sheet: /sound/

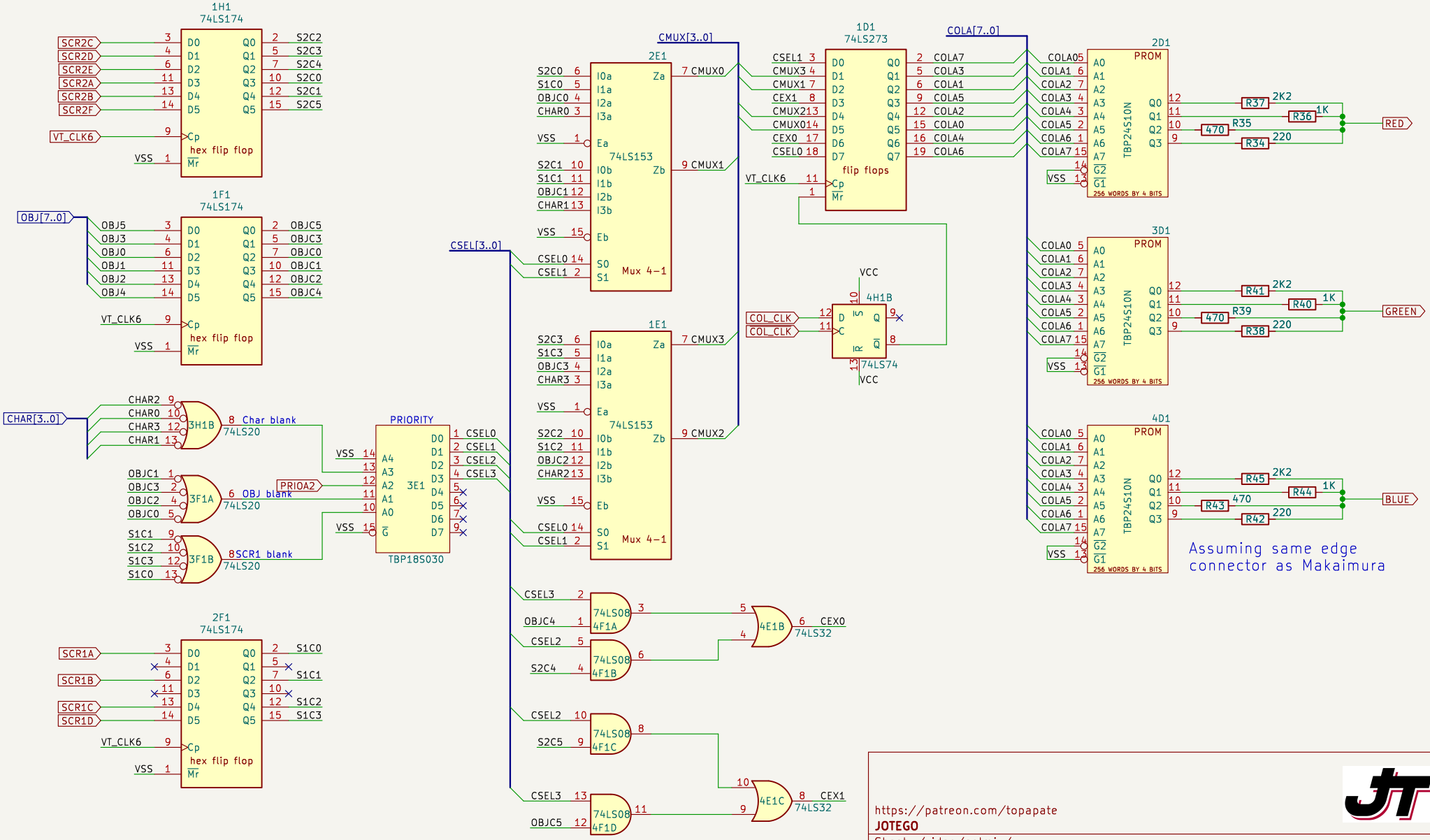
File: sound.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 11/28


E5
74LS138
address decoder
AB[6..0]
DB[7..0]
SD[7..0]
SDB[7..0]
VC2CS
CP137
301
VCC
VSS
SCR1X0CS
SCR1X1CS
SCR1Y0CS
SCR1Y1CS
SCR2X0CS
SCR2X1CS
PALBANKCS
GFXONCS
7B1
IO2CS
VCTRL1CS
RQB_CLK
SLATCHCS
Sound command latch
9F1
3-state flip flops
74LS374
SND_RQB
RQB
OT_RQB
74LS74
10F1
buffers
74LS245
SND_WAIT
10H1
SN76489AN
Sound Generator
R50
4.7kΩ
SG_RDY1
SRAMWE
PSG1RD
SG_CLK
CP153
10µF
R32
10kΩ
LPF
1/2πRC= 1.6kHz
R49
10Ω
CP158
103
11F1
R47
4.7kΩ
SG_RDY2
PSG0RD
Up to 4MHz
CP152
R33
R48
CP157
R52
220KΩ
PST518
Q_NPN_CEB
R51
47Ω
CP159
9C1
74LS367
3-StateHexBuffers
SINT
PORB
SWR
H0
SICLK
SRAMWE
SRST
SCLK
SRQB
CPU_RST
SG_CLK
PSG_CLK
74HC74
SNCS
4E1A
74LS32
https://patreon.com/topapate
JOTEGO
Sheet: /sound/snd_c_latch/
File: snd_c_latch.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
Rev: José Tejada
KiCad E.D.A. 8.0.9
Id: 12/28

COLOR PROMS



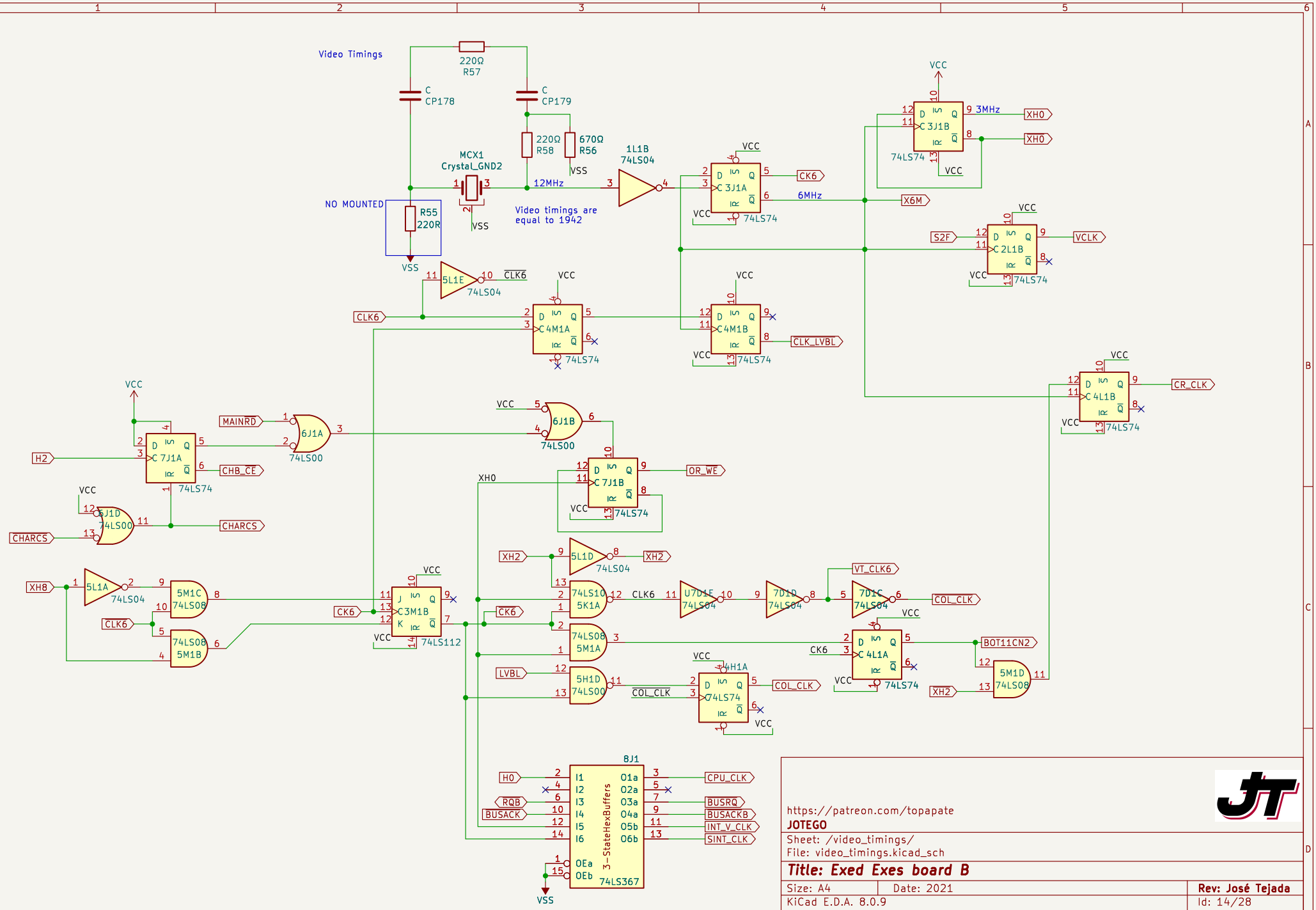
https://patreon.com/topapate

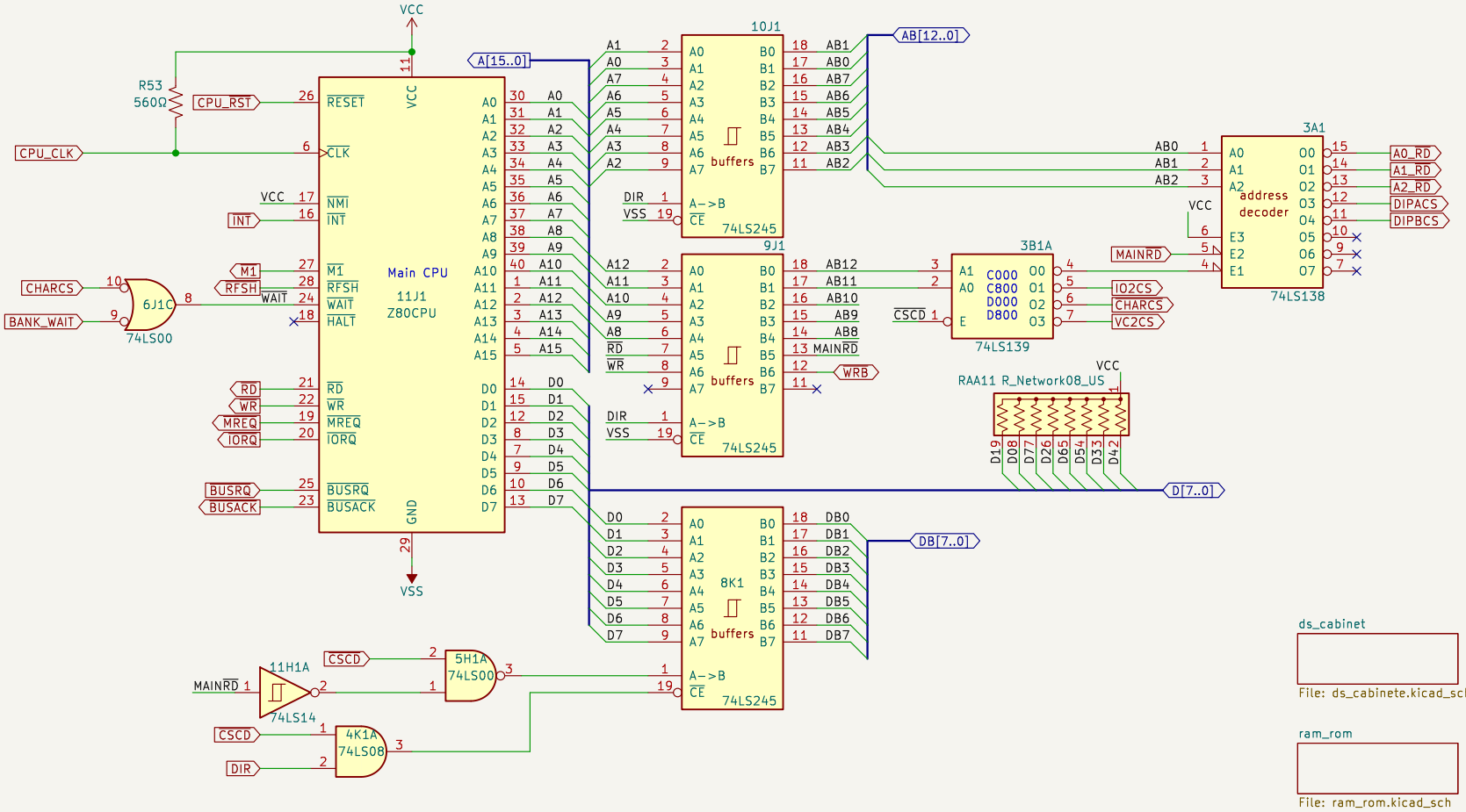
JOTEGO

Sheet: /video/colmix/
File: colmix.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 13/28


Video Timings
220Ω
R57
C
CP178
C
CP179
220Ω
R58
670Ω
R56
VSS
MCX1
Crystal_GND2
12MHz
NO MOUNTED
R55
220R
Video timings are
equal to 1942
1L1B
74LS04
VCC
3MHz
XH0
3J1B
74LS74
CK6
6MHz
X6M
S2F
2L1B
VCLK
5L1E
CLK6
4M1A
CLK_LVBL
4M1B
CR_CLK
4L1B
MAINRD
6J1A
74LS00
H2
7J1A
CHB_CE
6J1B
OR_WE
7J1B
6J1D
CHARCS
XH8
5L1A
74LS04
5M1C
74LS08
5M1B
3M1B
74LS112
XH2
5L1D
74LS10
5K1A
CLK6
U7D1E
74LS04
7D1D
7D1C
VT_CLK6
COL_CLK
74LS08
5M1A
4L1A
BOT11CN2
5M1D
LVBL
5H1D
74LS00
COL_CLK
4H1A
74LS74
BJ1
H0
RQB
BUSACK
3-StateHexBuffers
74LS367
CPU_CLK
BUSRQ
BUSACKB
INT_V_CLK
SINT_CLK
https://patreon.com/topapate
JOTEGO
Sheet: /video_timings/
File: video_timings.kicad_sch
Title: Exed Exes board B
Size: A4
Date: 2021
KiCad E.D.A. 8.0.9
Rev: José Tejada
Id: 14/28



ds_cabinet

File: ds_cabinete.kicad_sch

ram_rom

File: ram_rom.kicad_sch

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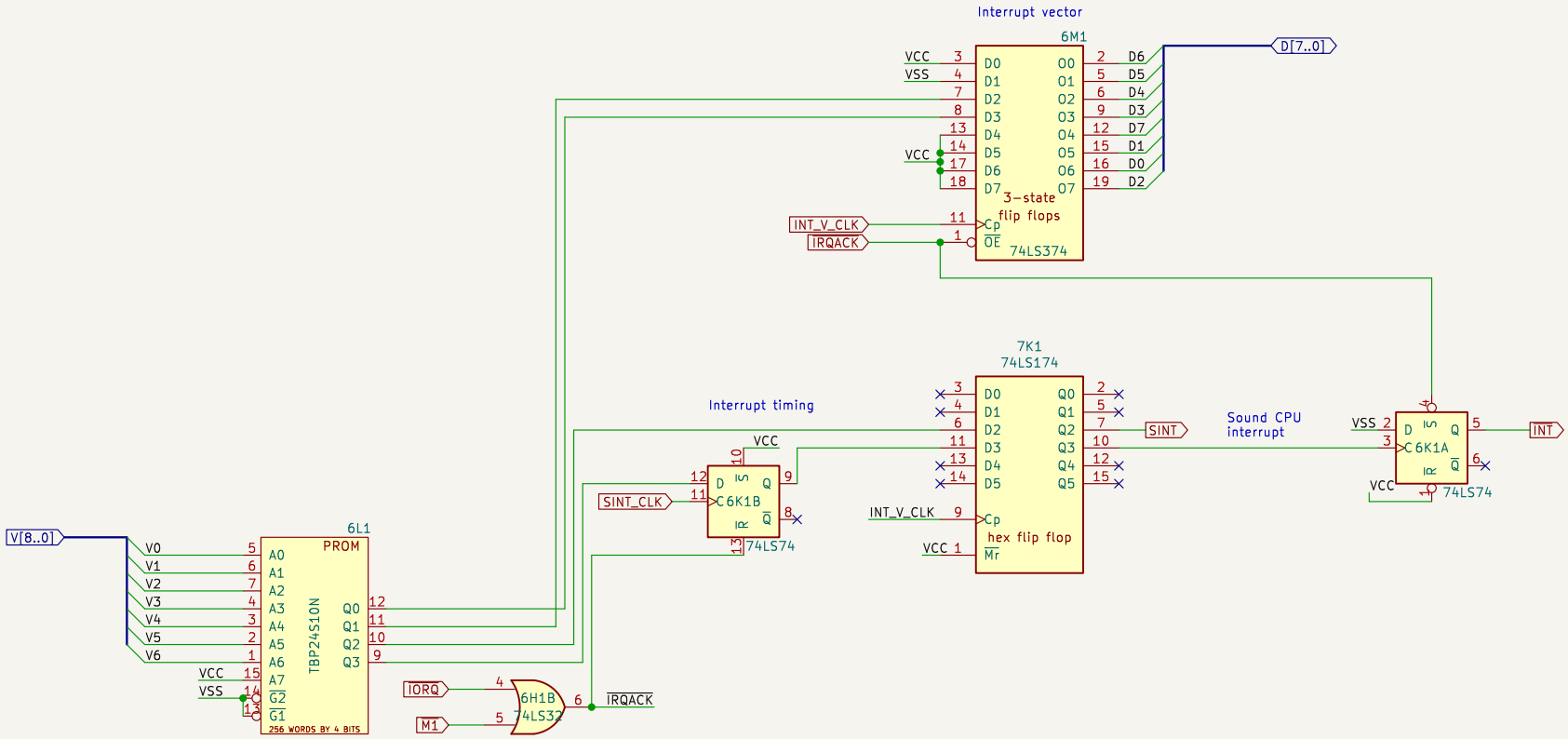
JOTEGO

Sheet: /main/

File: main.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 15/28



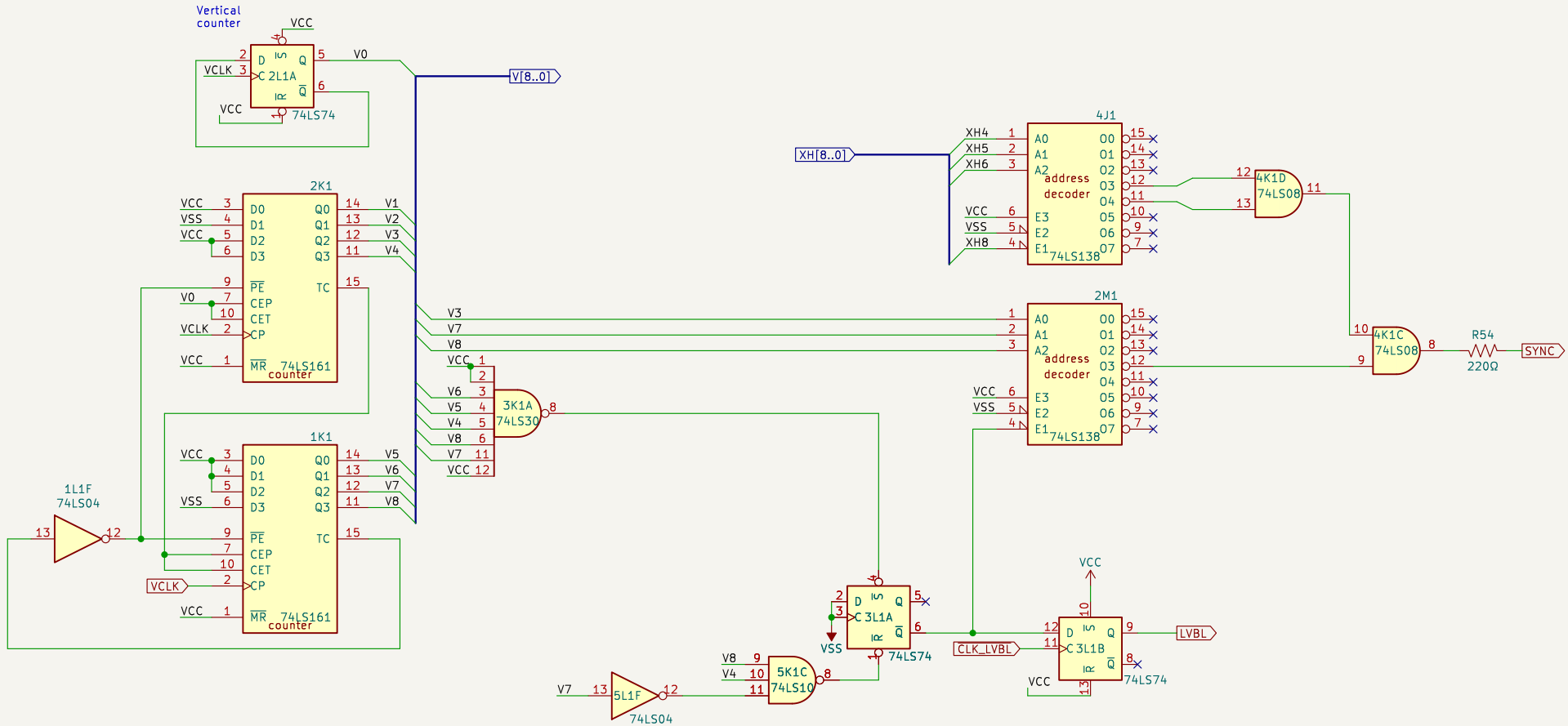
https://patreon.com/topapate

JOTEGO

Sheet: /interruption/
File: interruption.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 16/28



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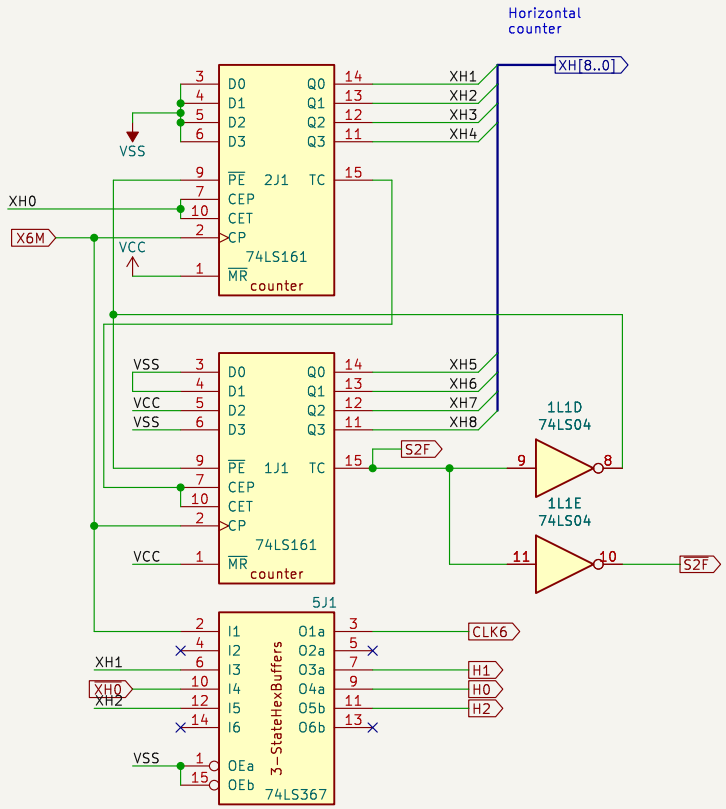
JOTEGO

Sheet: /v_counter/

File: v_counter.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	Rev: José Tejada
KiCad E.D.A. 8.0.9		Id: 17/28



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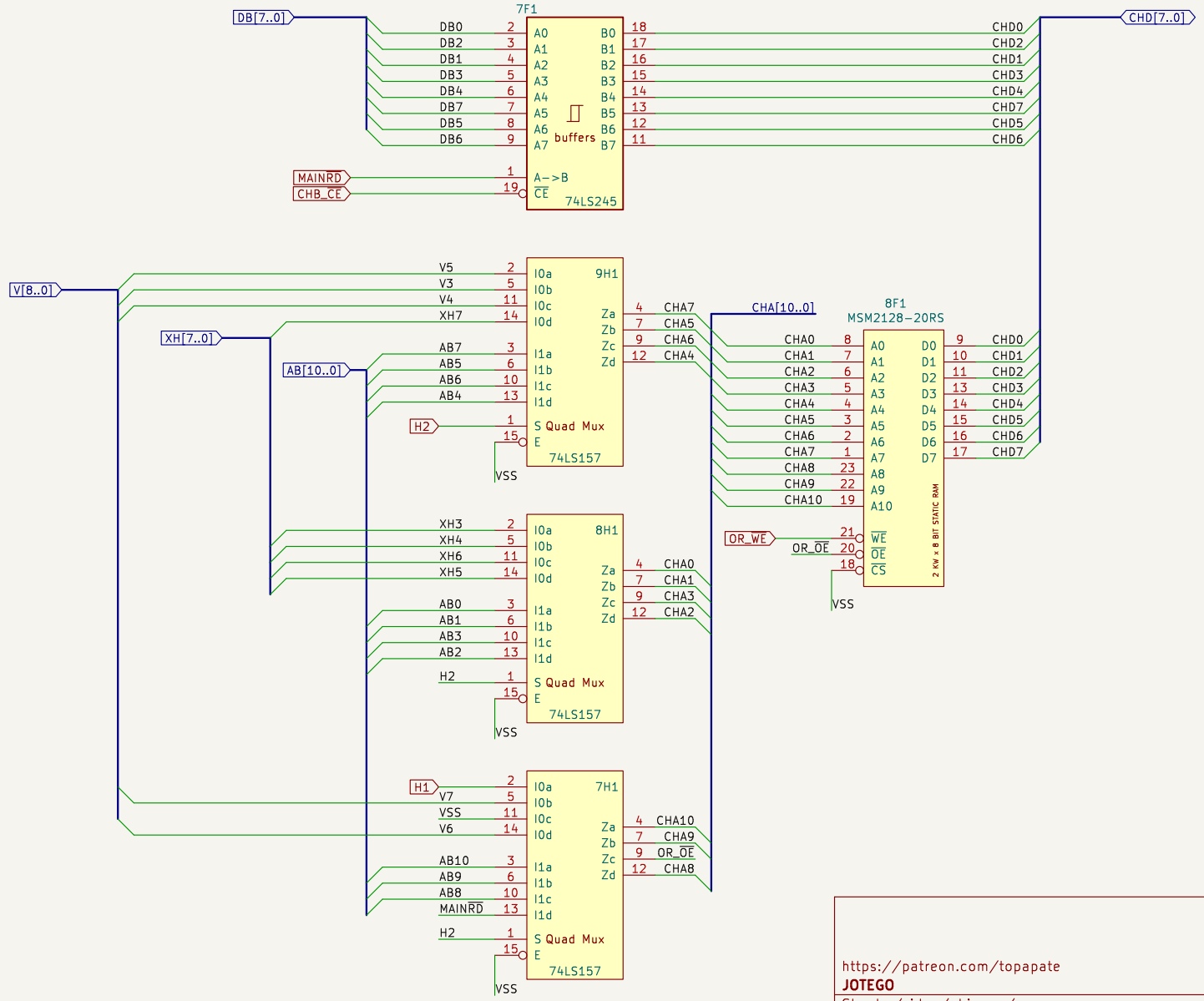
JOTEGO

Sheet: /h_counter/

File: h_counter.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 18/28



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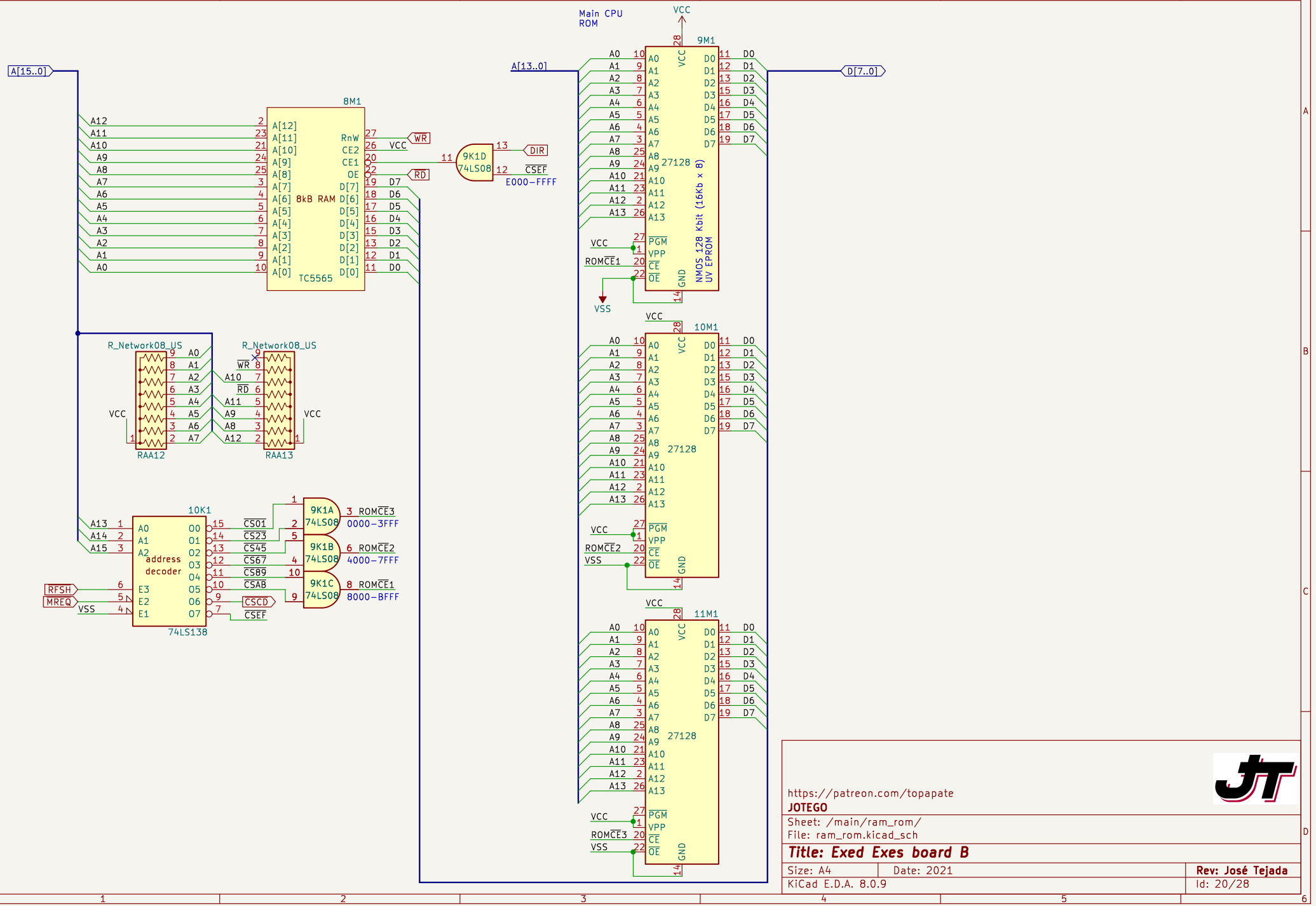
JOTEGO

Sheet: /video/obj_ram/

File: obj_ram.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 19/28



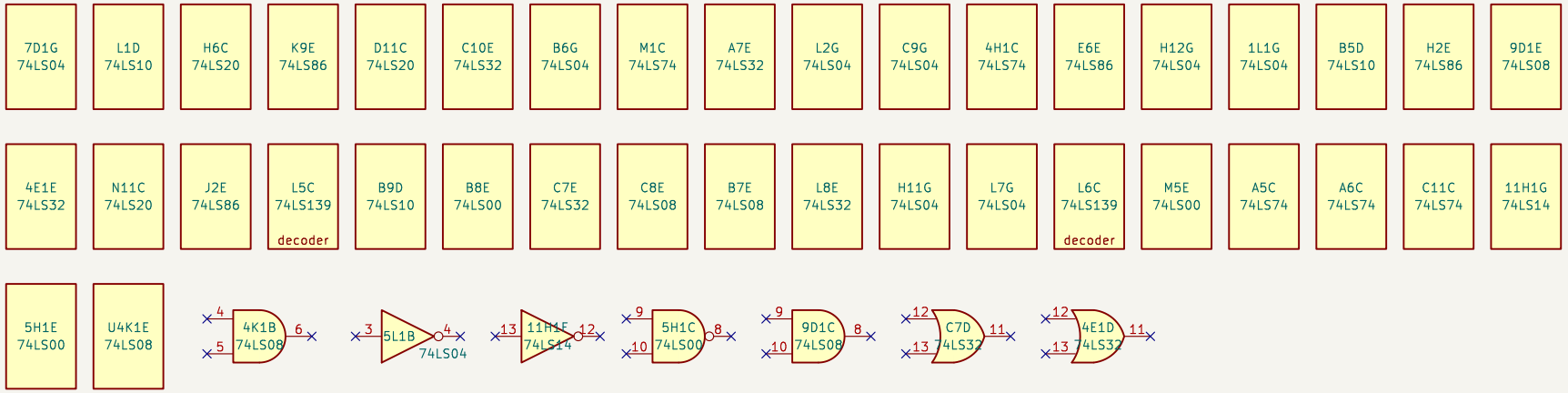
https://patreon.com/topapate

JOTEGO

Sheet: /main/ram_rom/
File: ram_rom.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 20/28



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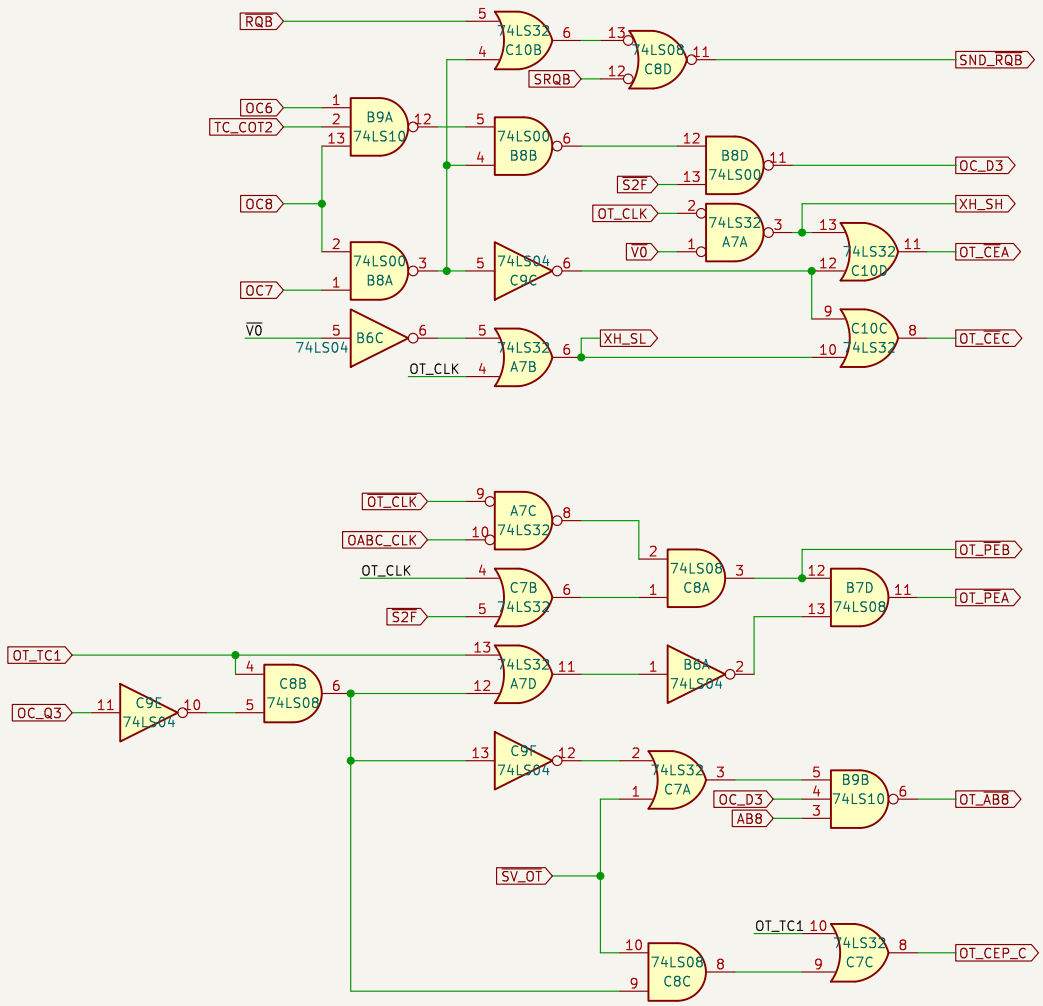
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Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 20/28



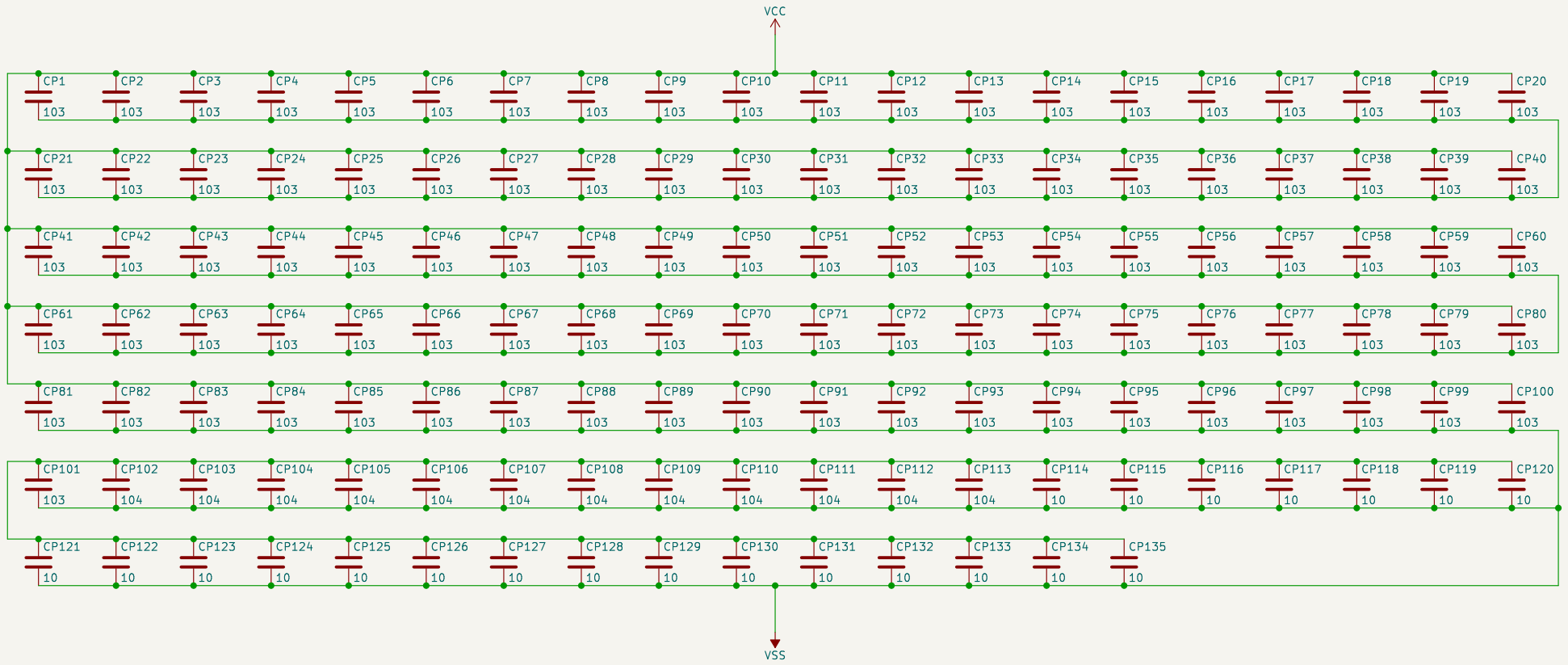
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Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 21/28



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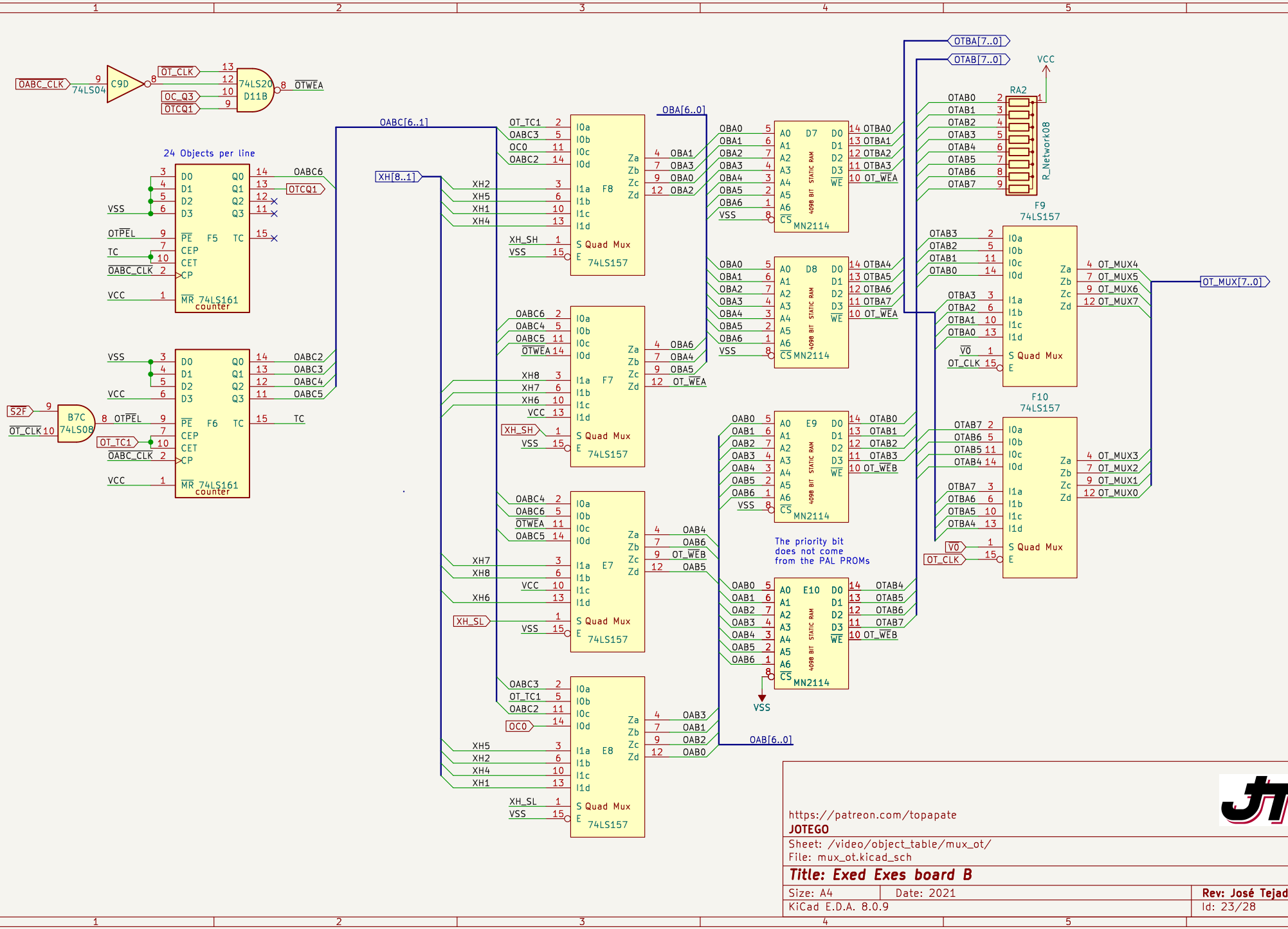
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Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 22/28



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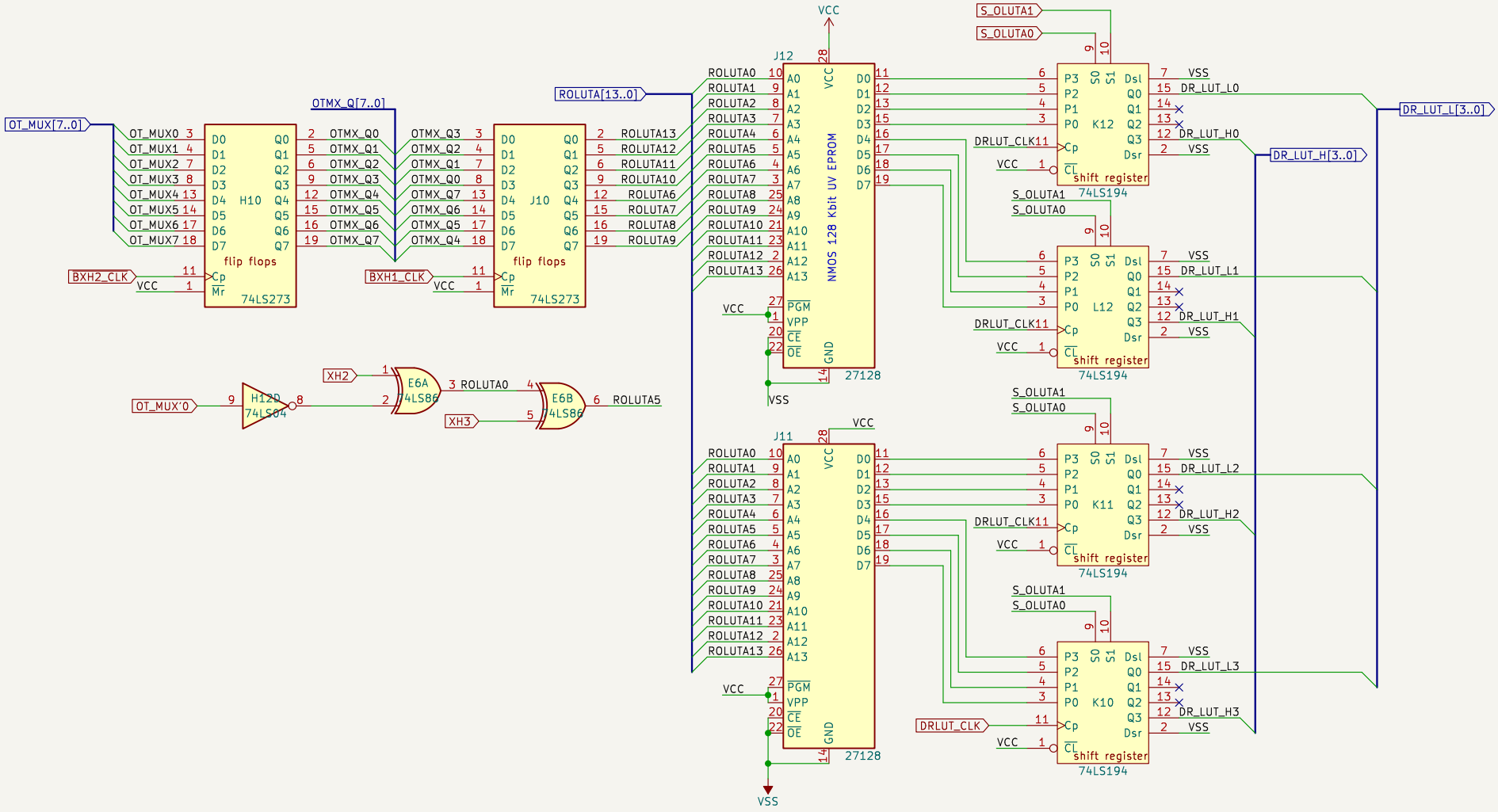
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Title: Exed Exes board B

Size: A4	Date: 2021	Rev: José Tejada
KiCad E.D.A. 8.0.9		Id: 23/28



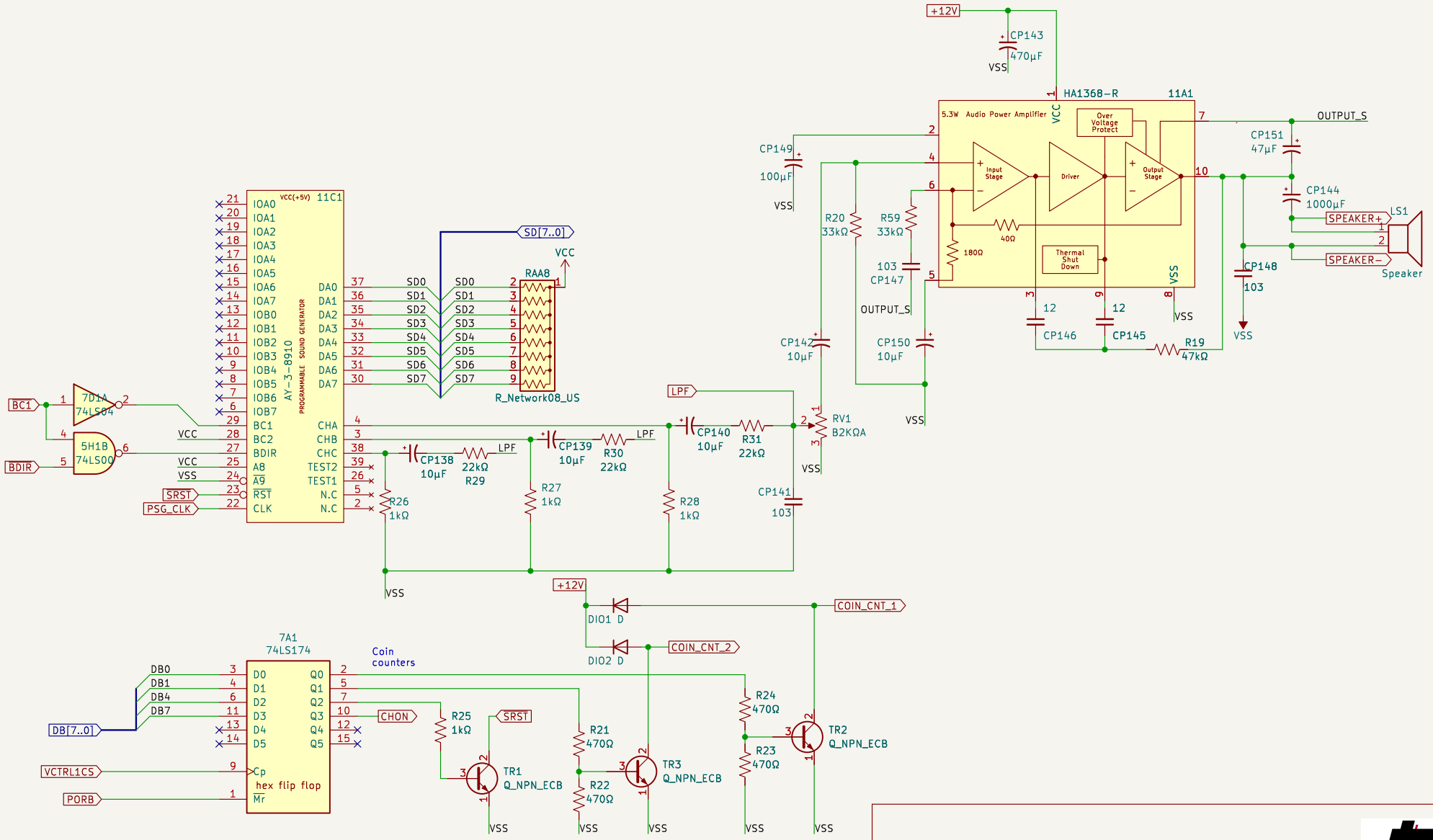
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JOTEGO

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Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 24/28



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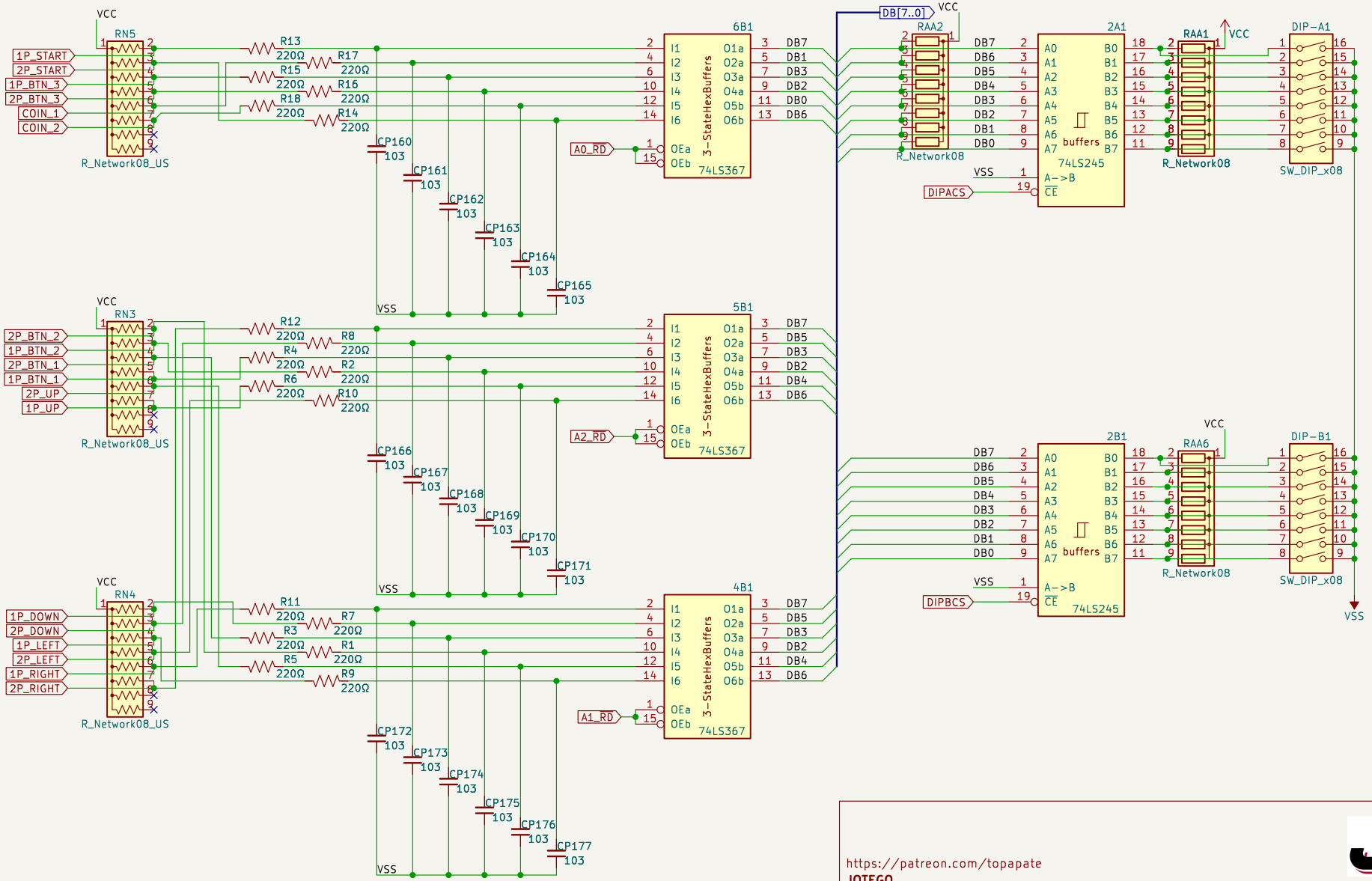
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File: psg.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 26/28



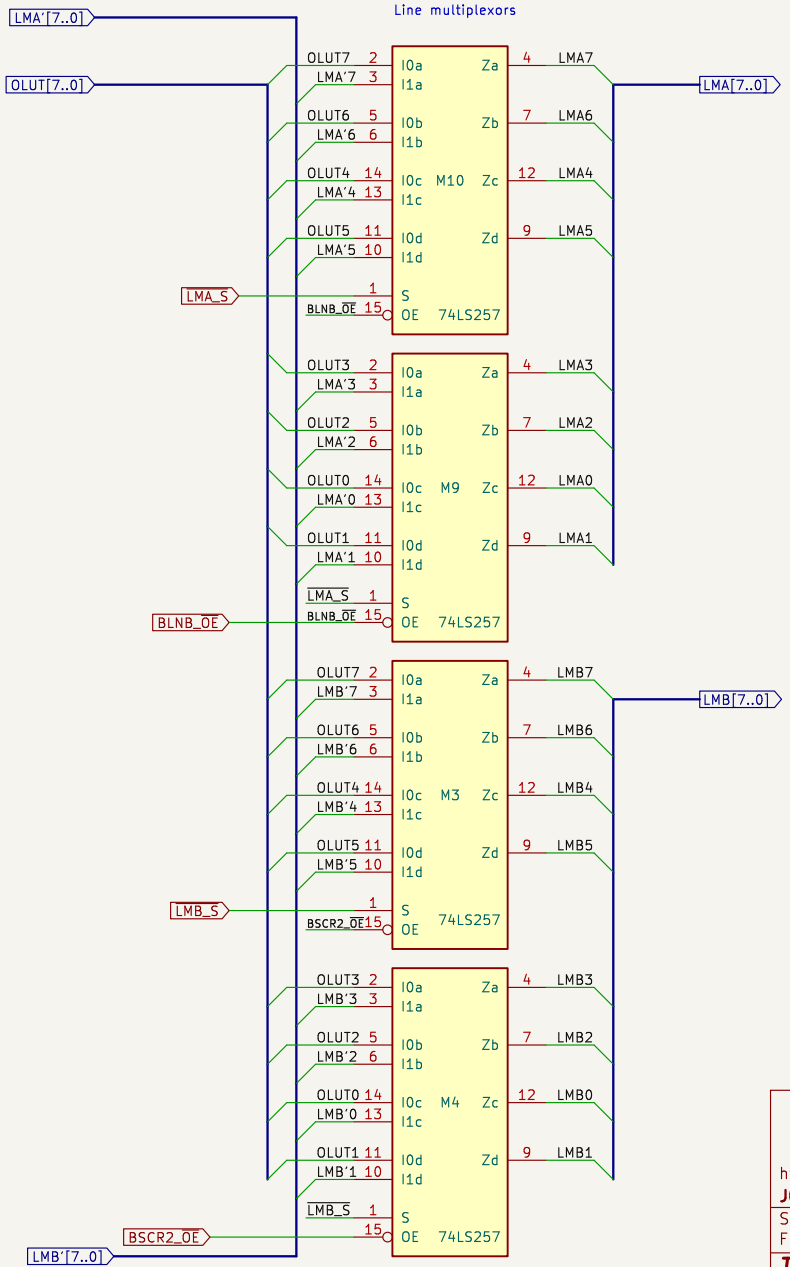
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JOTEGO

Sheet: /main/ds_cabinet/
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Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 26/28



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Sheet: /video/obj_counters/mux/
File: mux.kicad_sch

Title: Exed Exes board B

Size: A4	Date: 2021	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 28/28